

Sem 2 AY2122 MH1101 Calculus II Test (1 hour)

Name: Matriculation Number:

Tutorial Group:

Question 1 [16 marks]

(a) Calculate the derivative $\frac{d}{dx} \int_x^{x^3+\pi x} e^{\sin t} dt$. Express your final answer in terms of x .

(b) Write the following limit as a definite integral (do not evaluate it).

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{4i + n}{2i^2 + n^2}$$

Answer.

More space for answers.

Question 2 [14 marks]

- (a) Let R be the region bounded by the curves $y^2 = 9x$, $x + y = 10$ and the y -axis. Compute the volume of the solid formed by revolving R about the line $y = 10$.
- (b) Describe a solid of revolution (by identifying the region on the xy -plane to be rotated, and the axis of rotation) whose volume is given by the definite integral

$$\int_0^5 2\pi(5 - y)^2 dy.$$

Answer.

More space for answers.

More space for answers.

Question 3 [20 marks]

(a) Evaluate the following integrals:

(i) $\int \tan^2(3x) \sec^6(3x) dx$

(ii) $\int e^{2x} \sqrt{5 + 2e^x} dx$

(iii) $\int \sec^3 x dx$

(b) Assume that the following integrals exist. Determine whether the statement/equation below is (always) true or false. Justify your answer. Hint: Use substitution.

$$\int_0^\infty \frac{x^2}{x^4 + 1} dx = \frac{1}{2} \int_0^\infty \frac{x^2 + 1}{x^4 + 1} dx.$$

Answer.

More space for answers.

More space for answers.

More space for answers.

Formulae Table

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int k dx = kx + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \tan x dx = \ln |\sec x| + C$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1} x$$

$$\int e^x dx = e^x + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \cos x dx = \sin x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \cot^2 x = \csc^2 x$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\sin 2x = 2 \sin x \cos x$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\sin A \cos B = \frac{1}{2}(\sin(A - B) + \sin(A + B))$$

$$\sin A \sin B = \frac{1}{2}(\cos(A - B) - \cos(A + B))$$

$$\cos A \cos B = \frac{1}{2}(\cos(A - B) + \cos(A + B))$$

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

– End of Test –